

Face Mask Detection Using Image Processing

1) Introduction and Objective

By wearing face masks we can minimize the spreading of various diseases among people. It is not easy to manually check whether people are wearing masks or not in crowded areas like hospitals, schools and public transport. This is where Face Mask Detection Using Image Processing becomes very useful.

Project objective - to create an image processing system that helps to detect automatically whether someone is wearing a mask or not. I used image processing techniques with machine learning classifier to do this project.

For this project I used a dataset from Kaggle which is a well balanced one suitable for this.

Sample Input Images

With Mask



Without Mask



2) Methodology

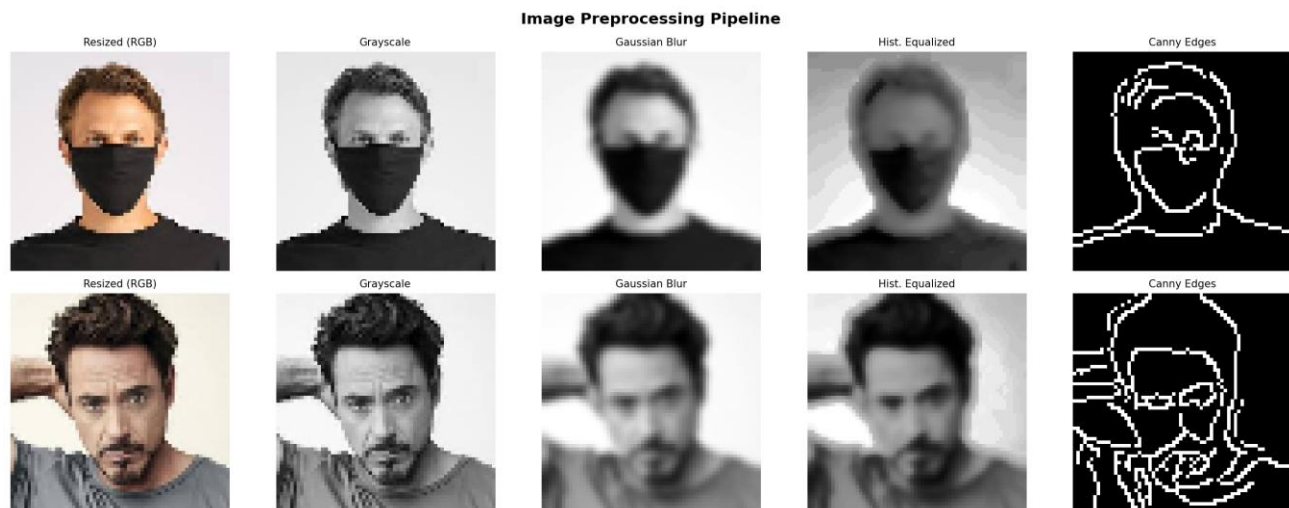
Image processing pipeline has four main stages in this project as - preprocessing, feature extraction, classification and evaluation.

2.1 Image Preprocessing Pipeline

Before extracting features all images go through a five-step preprocessing pipeline to standardize them and remove noise. Each step is applied consistently to all 7553 images:

- **Resize to 64 by 64 pixels:** All images are resized to a set size so that features can be compared regularly, not a issue is image big or small.
- **Grayscale conversion** - Color images are converted to grayscale. This removes color information and reduces the complexity of the data.
- **Gaussian Blur** - It is usedd to smooth the image and reduce high frequency noise that will be a issue in edge detection.
- **Histogram Equalization** - This helps to make features more visible in low light images by increasing the contrast of the image by redistributing pixel intensity values

- Canny Edge Detection - Edges are detected using the algorithm which identifies sharp intensity changes in the image highlighting structural features like the face mask boundary.



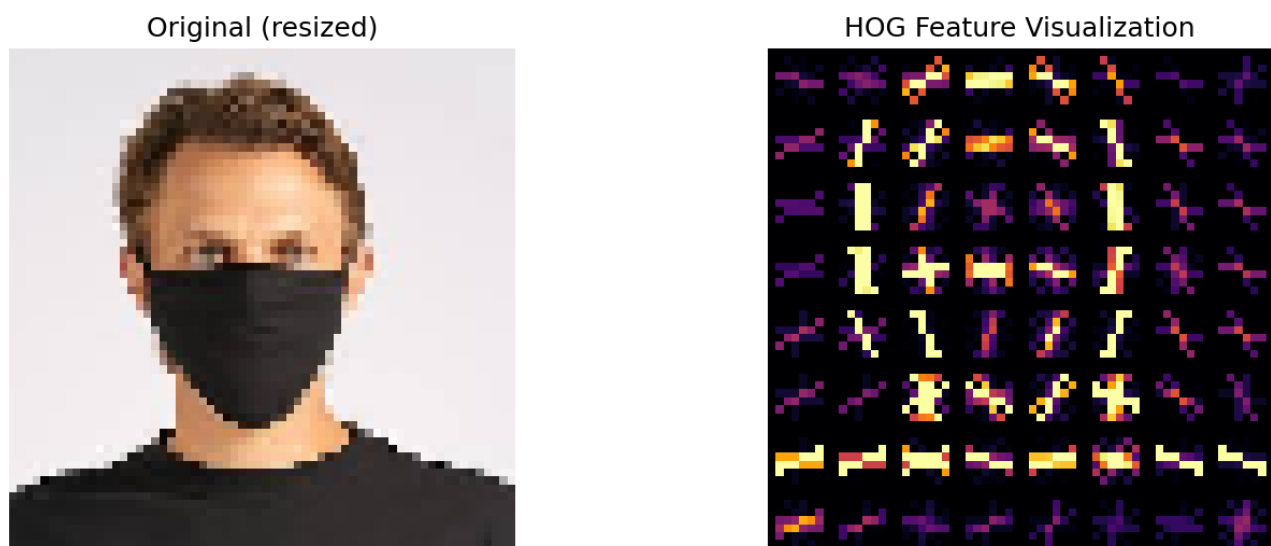
2.2 Feature Extraction

Feature extraction is the important step in the pipeline. Of feeding raw pixel values into the classifier we extract two types of meaningful features from each image and combine them into a single feature vector of 1664 dimensions.

HOG features capture the shape and texture of the face region. HOG works by dividing the image into cells and computing the distribution of gradient directions in each cell. Since a face mask significantly changes the shape of the face HOG features are very effective for this task.

Colour Histogram features capture the color distribution of the image. A face mask introduces a color pattern in the nose and mouth region. For the each 3 colour channels if we compute a histogram and then by normalizing we will be able to get a representation of the color information

HOG Feature Extraction



2.3 Classification - SVM

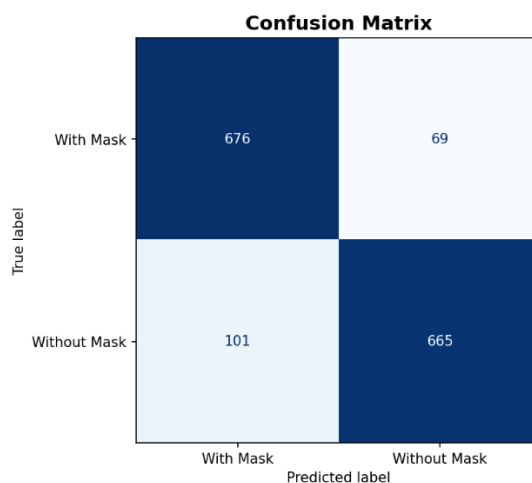
I picked a support vector machine with an RBF kernel as the classifier. Support Vector Machine is a good choice for this kind of classification task because it can find the best decision boundary between two classes.

3) Results

3.1 Confusion Matrix

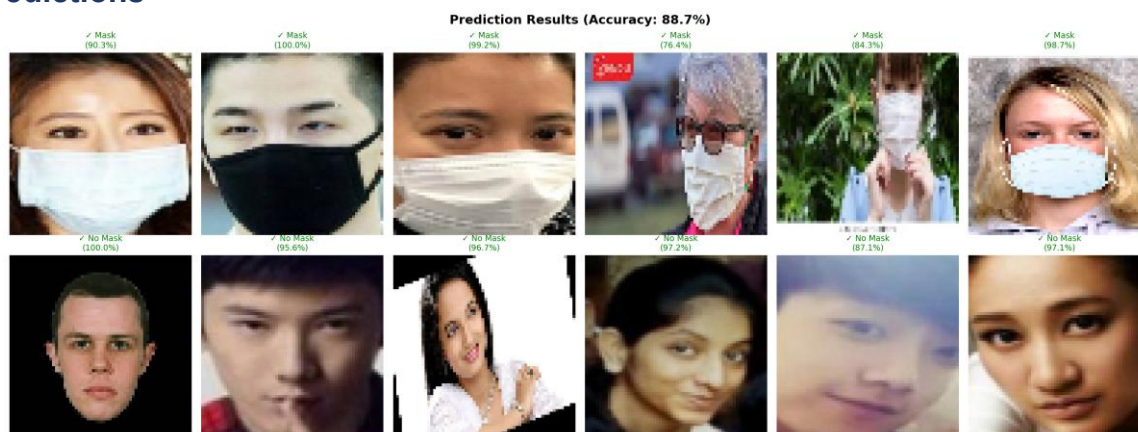
The trained Support Vector Machine model was tested on 1511 images that it had never seen before. The model got 88.75% of the tests right.

The confusion matrix shows us how the predictions were broken down. There were 745 images of people wearing face masks in the test set. Of those, 676 were correctly classified, and 69 were wrongly predicted to be without face masks. There were 766 images of people without face masks, and 665 of them were correctly identified. 101 of them were incorrectly identified as wearing masks. We can see below



This shows that both classes got an F1-score of about 0.89, with precision and recall balanced at 87-91%. This shows that the model works the same for both masked and unmasked faces.

3.2 Predictions



4) Discussion

4.1 Challenges

The biggest challenge was in some images there was high lighting and the images are taken in high angles so because of it was bit difficult to extract clean features and if the person in image not wearing the mask properly or wearing stuff like sunglasses also caused some errors during classification.

4.2 Insights and Observations

One key observation was that the Canny edge detection step clearly shows a difference between face mask faces and non-face mask faces. For the face mask face there is a strong horizontal edge at the top of the face mask that is completely absent in the non-face mask face.

The HOG visualization confirms that the gradient activity is concentrated around the face mask region in face mask images along the face mask edges and the nose bridge. This pattern is consistently different from the distribution in non-face mask faces, which explains why HOG features work well here.

4.3 Limitations and Possible Improvements

The main limitation of the system is that it classifies the entire image as face mask or non-face mask. It does not detect where the face mask is located or whether the face mask is worn correctly. Another limitation is the 64 by 64 resize which loses a lot of detail from the image.

We can improve this project by having face detection before doing classification by that we will be able to increase the accuracy of project and also we can improve the project by having live video capture system for real time detection of face masks prescene.

5) Conclusion

This project built a face mask detection system using image processing techniques and a machine learning classifier. The system reached a test accuracy of 88.75% on a dataset of 7553 face images. It used a Support Vector Machine classifier along with combined HOG and color histogram features.

The project demonstrates that traditional image processing methods, without learning can still achieve strong results for binary classification tasks when the right features are selected. The approach is also computationally efficient. Can run without a GPU. With improvements such as face detection preprocessing and CNN-based feature extraction the accuracy could be pushed even higher for real-world deployment.